

U-MV: Regional-scale *Acacia tortilis* crown mapping from UAV imagery

Technical hand-over and reproduction guide

Software, environment, data, training, evaluation and regional inference

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Prepared by Mohamed Barakat A. Gibril

GIS and Remote Sensing Center, Research Institute of Sciences and Engineering, University of Sharjah

Contact: mbgibril@sharjah.ac.ae

The dataset described herein is not publicly shareable and is provided for project use only.

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1 Read this first

This document accompanies the hand-over of the U-MV (U-shaped MambaVision) framework for *Acacia tortilis* crown mapping. It is written for a team member who has (i) the public code repository and (ii) read access to the project archive `A.tortilis_Data & Model` on the shared drive, which holds the dataset and the original MMSegmentation work directories with the trained checkpoints. Following it, that person installs the software, verifies it, reproduces the published accuracy figures and produces regional crown maps from new UAV orthomosaics without further assistance.

1.1 What you receive

Item	Where	What it is
Code	https://github.com/brakuta/U-MV-Acacia-tortilis-Crown-Mapping	model, configs, Docker build, tools, tests, this guide
Dataset	<code>Z:\...\A.tortilis_Data & Model\Data</code> used to build the model	<code>img_dir/</code> and <code>ann_dir/</code> with <code>train</code> , <code>val</code> , <code>test2</code> , <code>Generalizability</code> (1024×1024 tiles, ~32 GB)
Trained models	<code>Z:\...\A.tortilis_Data & Model\A.tortilis_Models\Pretrained weights</code>	three MMSegmentation work directories with checkpoints, training configs and logs (~2.5 GB)

The archive also contains `A.tortilis_Models\MMSegmentation_Folder`; it is the superseded container workspace and is **not** needed.

1.2 How to use this document

- **Chapter 2 is the procedure.** It is written for a first-time user of command windows, Git and Docker: step 0 explains the words used, steps 1 to 14 take an empty Windows workstation to the reproduced test metrics and a mapped orthomosaic, and the table "If you see this message" at its end covers the errors observed so far. Each step names where its commands run, gives them, and states the expected result. Work through chapter 2 alone; consult the rest only when a step refers to it.
- Chapter 3 describes the framework. Chapters 4 to 8 are the reference behind the steps: installation, data, training, evaluation and regional inference, including the Linux/WSL2 forms of every command.
- Chapter 9 maps the paper to the configuration files and records the reconciliation with the original training logs. Chapter 10 lists failure modes with remedies.
- The appendices record what changed in the code revision of September 2026, describe the released weights, and give a one-page command reference.

1.3 Conventions

PowerShell commands run in a Windows PowerShell window in the repository folder (`D:\U-MV`). Commands marked "inside the container" run in the Linux shell started by step 8, where the dataset is mounted at `/data` and the checkpoint folder at `/weights`. Paths on the host are examples and are adapted where the procedure says so.

2 Setup procedure (Windows, from scratch)

This chapter is the complete procedure for the receiving team member. It assumes no previous experience with command windows, Git or Docker, and it starts from an empty Windows 11 workstation with an NVIDIA GPU (12 GB of GPU memory is enough), 64 GB of RAM, and read access to the project share (Z:). Every command is copied exactly as printed and pasted into the window with Ctrl+V, followed by Enter. Each step says which window it runs in, shows the commands, states what a successful result looks like, and what to do otherwise. The table "If you see this message" at the end of the chapter covers every error observed so far. The other chapters are reference material and are not needed to complete this one.

Step	What	Where	Time
0	Words you will see	–	3 min
1	Install the driver, Docker Desktop and Git	Windows	20 min
2	Download the code	PowerShell	1 min
3	Copy the archive from Z: to the local disk	PowerShell	10–60 min
4	Give Docker enough memory	PowerShell	3 min
5	Confirm Docker can use the GPU	PowerShell	1 min
6	Check that Docker knows where the data is	PowerShell	1 min
7	Build the image	PowerShell	15–40 min
8	Start the container	PowerShell	1 min
9	Verify the software	container	3 min
10	Verify the dataset	container	3 min
11	Identify the checkpoint	container	1 min
12	Reproduce the published accuracy	container	40–60 min
13	Map an orthomosaic (optional)	container	varies
14	Leave and come back	both	–

Steps 1 to 7 are done once. Steps 9 to 12 verify the installation. Facts to know before starting:

- **Which weights.** `best_mIoU_iter_*.pth` in each work directory (the checkpoint with the best validation mIoU, used for the paper's test results): tiny `best_mIoU_iter_100000.pth`, small `best_mIoU_iter_95000.pth`, base `best_mIoU_iter_60000.pth`. Every tool accepts the folder instead of the file and selects it automatically.
- **Which model.** Start with U-MV-small; it produced the regional maps and has the best generalisability figures.
- **Dataset.** The archive holds 4 893 (train), 2 407 (val), 3 123 (test2) and 2 162 (Generalizability) tiles of 1024×1024 pixels. The published models were trained on 26 615 tiles; the training folder was pruned afterwards. Evaluation and inference are exact; retraining reproduces the recipe, not the model.
- **Confidentiality.** The dataset is not publicly shareable.

Step 0 — Words you will see

- **PowerShell** is the window in which commands are typed. Open it with the Windows key, type PowerShell, click **Windows PowerShell** (blue icon). Never choose "Run as administrator": an administrator window cannot see the Z: drive.
- **Prompt** is the line where you type. `PS C:\Users\name>` or `PS D:\U-MV>` means Windows PowerShell. `root@1a2b3c4d: /workspace/U-MV#` (the letters and digits differ every time) means the Linux container of

step 8. Each step says which prompt it needs; a command typed at the wrong prompt fails with "is not recognized" or "command not found".

- **Run a command:** click after the prompt, paste with Ctrl+V (or right-click), press Enter. Nothing happens until Enter is pressed. If Windows asks "You are about to paste text that contains multiple lines", click **Paste anyway**.
- **Finished** means a new prompt line with a blinking cursor has appeared. Until then, wait, even if nothing is printed for thirty minutes, and do not close the window. Ctrl+C stops a command that seems stuck.
- **Copy exactly.** Every quote ", backslash \, slash /, dash - and dollar sign \$ matters; do not add or remove any. Folder names used here must not contain the character &.
- **Repository** is the folder with the program's code, D:\U-MV, created in step 2. `cd D:\U-MV` moves the prompt into it; it is the first line of every PowerShell step.
- **Docker Desktop** is a program that runs a ready-made Linux system. The **image** is that system, built once in step 7. The **container** is the image running in your window (step 8); typing `exit` leaves it.
- **Mounted folder:** a folder of your disk is visible inside the container under a Linux name. D:\A.tortilis_Data_Model\Data used to build the model is `/data` there, and the trained models are `/weights`.
- **Windows questions.** "Do you want to allow this app to make changes?" appears only for the installers of step 1: answer **Yes**.
- **Red text** is an error. Read only its first line, find it in the table "If you see this message" at the end of the chapter, do what the table says, and do not continue to the next step until the current one succeeds.

Step 1 — Install the driver, Docker Desktop and Git

Windows · once per machine · needs Internet

1a. Graphics card and driver. Open PowerShell (see step 0) and paste both lines:

```
Get-CimInstance Win32_VideoController | ForEach-Object { $_.Name }
nvidia-smi --query-gpu=name,memory.total,driver_version --format=csv
```

You should see an NVIDIA card in the first answer, and then two lines such as `name, memory.total [MiB], driver_version` and `NVIDIA GeForce RTX 3060, 12288 MiB, 560.94`. The driver number must be 520 or higher.

If the first answer names no NVIDIA card (only Intel or AMD), this computer cannot run the models: report it and stop here. If an NVIDIA card is listed but the second command answers The term 'nvidia-smi' is not recognized, the driver is missing: install it for that card from nvidia.com, restart Windows and repeat 1a.

1b. Docker Desktop. Download "Docker Desktop for Windows" from docker.com and run the installer (answer **Yes**; keep "Use WSL 2 instead of Hyper-V" ticked; restart the computer when asked). Start Docker Desktop from the Start menu. On first start click **Accept** on the agreement and **Skip** or **Continue without signing in** on the sign-in page; no account is needed. If it says that WSL must be installed or updated, paste `wsl --update` into PowerShell, restart the computer and start Docker Desktop again. Docker is ready when the whale icon in the taskbar corner (click the small ^ arrow to see hidden icons) has stopped moving. Then check in PowerShell:

```
docker version
```

You should see a `Client:` block and a `Server: Docker Desktop` block. If the command is "not recognized", close PowerShell and open a new window (programs installed while a window is open are not visible in it).

1c. The Linux layer. Docker runs the software inside a small Linux system built into Windows, called WSL 2. Without it the graphics card stays invisible to Docker, whatever else is done. Paste both lines:

```
wsl --version
wsl --update
```

You should see a block naming a WSL version and a kernel version, then either The requested operation is successful or a message that it is already up to date.

If the answer is The term 'wsl' is not recognized, WSL is not installed on this computer, and it must be installed before anything else works. It needs administrator rights and two restarts, so it is usually a task for the computing support team. In a PowerShell started with **Run as administrator**:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

Restart Windows, then in a normal PowerShell paste `wsl --install`, restart Windows again, and repeat 1c. On a company machine these features may be blocked by policy; in that case the computing support team must enable them.

1d. Git. In PowerShell paste `git --version`. If it prints `git version 2.x`, Git is installed. Otherwise install "Git for Windows" from git-scm.com with all default options, close PowerShell and open a new window.

1e. Disk space. Paste:

```
Get-PSDrive C,D | ForEach-Object { "{0}: {1} GB free" -f $_.Name, [math]::Round($_.Free/1GB) }
```

You should see two lines such as C: 120 GB free and D: 800 GB free; 40 GB on C: (the image) and 40 GB on D: (the archive) are needed. If the answer contains Cannot find drive for D, the computer has no D: drive: use C: instead and replace D:\ by C:\ in every command of steps 2, 3 and 6.

1f. Docker must use the Linux layer. Paste:

```
docker info --format "{{.KernelVersion}}"
```

You should see a version ending in `-microsoft-standard-WSL2`. *If it ends in -linuxkit*, Docker Desktop is running on Hyper-V instead, where the graphics card is never visible: open Docker Desktop → Settings → General → tick **Use the WSL 2 based engine** → **Apply & restart**, wait until the whale icon is still, and repeat 1f. If that box cannot be ticked, WSL is missing: go back to 1c.

Step 2 — Download the code

PowerShell · once · needs Internet · no GitHub account is needed

The second line removes a folder left by an earlier attempt (it stays silent if there is none). The third stops Git from downloading the large weight files stored on GitHub: they are not needed, the archive already has them, and the download can fail with a quota error.

```
cd D:\
Remove-Item -Recurse -Force D:\U-MV -ErrorAction SilentlyContinue
$env:GIT_LFS_SKIP_SMUDGE = "1"
git clone https://github.com/brakuta/U-MV-Acacia-tortilis-Crown-Mapping.git U-MV
cd D:\U-MV
git log -1 --format=%cd
```

You should see Cloning into 'U-MV'..., Receiving objects: 100%, Resolving deltas: 100% ..., done. and finally a date. The prompt is now `PS D:\U-MV>`. The date must be recent; if it is older than the date printed on the front page of this guide, tell the project lead before continuing.

Step 3 — Put the dataset on the local disk

PowerShell, in D:\U-MV · once · 10 to 60 minutes if it must be copied

Tiles must not be read from the shared drive (too slow), so the dataset and the trained models have to be on a disk of this computer. The command of this step also checks them and writes the settings file that step 6 needs.

3a. Is the archive already on this computer? Paste:

```
cd D:\U-MV
$F = "Data used to build the model"
Get-ChildItem D:\ -Directory -Recurse -Depth 4 -Filter $F -EA 0 | ForEach-Object { $_.Parent.FullName }
```

This takes up to a minute. It prints either nothing, or one line per copy found: the folder that holds the archive, for example `D:\Vegetation\3_Mapping Acacia tortilis Trees\A.tortilis_Data_Model`. If several lines appear, use the first one in 3b; if its check reports CHECK, try the next.

3b. A line was printed: check that copy, copy nothing. Paste the folder between the quotes, exactly as printed, and keep the word check at the end:

```
tools\windows\copy_archive.cmd "D:\Vegetation\3_Mapping Acacia tortilis Trees\A.tortilis_Data_Model" check
```

3c. Nothing was printed: copy the archive from the share. This takes 10 to 60 minutes, and nothing new appears on the screen while it copies:

```
tools\windows\copy_archive.cmd D:\A.tortilis_Data_Model
```

You should see, after either command, four lines such as `train images= 4893 masks= 4893 ok` (also `val 2407, test2 3123, Generalizability 2162, each marked ok`), three `best_mIoU_iter_*.pth` paths under `Best checkpoints found:`, a line `Written: D:\U-MV\docker\.env`, and the last line `RESULT: OK`. After 3c there are also two robocopy tables, in which the `Files` : row must show 0 under FAILED.

If red text says Source not found, the shared drive is not connected: open File Explorer → This PC and check that Z: opens, make sure PowerShell was not started as administrator, then repeat. *If the folder name contains the character &*, rename it in File Explorer first (Docker cannot use it). *If the share has another drive letter* (for example Y:), use:

```
$src = "Y:\Final Geodatabase\Vegetation_Geodatabase\3_Mapping Acacia tortilis Trees"
tools\windows\copy_archive.cmd D:\A.tortilis_Data_Model -Source "$src\A.tortilis_Data & Model"
```

The command may be repeated at any time: it copies only what is missing.

Step 4 — Give Docker enough memory and restart it

PowerShell · once

Docker runs inside the Linux layer of Windows, whose memory limit is set in a small file. Three actions, in this order.

4a. Write the file (40 GB of the 64 GB; the second line shows the result):

```
$cfg = "$env:USERPROFILE\.wslconfig"
Set-Content $cfg -Encoding ASCII -Value '[wsl2]', 'memory=40GB', 'swap=8GB'
Get-Content $cfg
```

You should see the three lines `[wsl2], memory=40GB, swap=8GB`.

4b. Restart the Linux layer so that the file and the update of step 1c take effect. Right-click the whale icon in the taskbar corner (click the small ^ arrow if it is hidden) → **Quit Docker Desktop**, and wait until the icon has disappeared. Then paste:

```
wsl --shutdown
```

It prints nothing. Now start Docker Desktop again from the Start menu and wait until the whale icon is still. This takes up to two minutes.

4c. Check the result. Paste:

```
[math]::Round([long](docker info --format "{{.MemTotal}}") / 1GB)
```

You should see 39 or 40. A smaller number means Docker was not restarted after the file was written: repeat 4b. An error message means Docker Desktop is not running yet: wait for the whale icon and repeat 4c.

Step 5 — Confirm Docker can use the GPU

PowerShell · once · Docker Desktop must be running

```
cd D:\U-MV
docker\umv.cmd gpu
```

You should see, after a short download (Unable to find image ... locally and Pull complete lines are normal), a line with the GPU name, its memory, the driver version and a number such as 8.6, followed by OK: Docker can use the GPU shown above.

If it fails, the usual repair is, in this order: paste `wsl --update`, quit Docker Desktop from the whale icon, paste `wsl --shutdown`, start Docker Desktop again from the Start menu, and repeat step 5.

If it still fails, this command prints a report that names the cause. It only reads, it starts and changes nothing:

```
docker\umv.cmd doctor
```

It prints six sections: the Windows version, the NVIDIA driver, the Linux layer, the Docker engine, the driver inside the Linux layer, and a last attempt in a container. Each section states what it needs. The two decisive lines are section 2, which must show the graphics card and a driver version of 520 or higher, and section 4, whose kernel must contain WSL2; a kernel containing `linuxkit` means Docker Desktop runs on Hyper-V, where no graphics card is visible, and the remedy is Docker Desktop → Settings → General → tick **Use the WSL 2 based engine** → Apply & restart. Send the whole report to the project lead.

Do not stop the work meanwhile: **the graphics card is not needed for step 7**, so go on and build the image, which takes the longest of all steps. It is needed from step 8 onward.

Step 6 — Check that Docker knows where the data is

PowerShell, in D:\U-MV · once

Step 3 wrote this file; here it is only read back.

```
cd D:\U-MV
type docker\.env
```

You should see exactly three lines, the first two naming the folder found in step 3, for example

```
DATA_DIR="../../../A.tortilis_Data_Model/Data used to build the model"
WEIGHTS_DIR="../../../A.tortilis_Data_Model/A.tortilis Models/Pretrained weights"
HF_HUB_OFFLINE=0
```

where ... stands for the beginning of the folder found in step 3, written with forward slashes.

The slashes lean forward and each line ends with a double quote; that is correct. *If the file is not found* or the folder is wrong, repeat step 3 with the right folder, or write the file by hand: set `$d` to that folder with forward slashes and no slash at the end, then paste all five lines together:

```
$d = "D:/Vegetation/3_Mapping Acacia tortilis Trees/A.tortilis_Data_Model"
$l1 = 'DATA_DIR="' + $d + '/Data used to build the model"'
$l2 = 'WEIGHTS_DIR="' + $d + '/A.tortilis Models/Pretrained weights"'
Set-Content docker\.env -Encoding ASCII -Value $l1, $l2, 'HF_HUB_OFFLINE=0'
type docker\.env
```

Step 7 — Build the image

PowerShell · once · 15 to 40 minutes · needs Internet

```
cd D:\U-MV
docker\umv.cmd build
```

Leave the window open and the computer awake (Settings → System → Power: never sleep when plugged in). Lines starting with # followed by a number show the stages: base image download, apt packages, conda packages, the mmcv compilation (several minutes with little output), the project packages, mamba_ssm OK, umv 1.1.0.

You should see the last line Build finished. Next: docker\umv.cmd shell.

If it ends with BUILD FAILED, paste the same two lines again once; downloads are often interrupted and finished stages are reused. If it fails a second time, click the window's title bar with the right mouse button → Edit → Select All, then Enter (this copies the whole window), paste into an e-mail to the project lead.

Step 8 — Start the container

PowerShell · every session · Docker Desktop must be running

```
cd D:\U-MV
docker\umv.cmd shell
```

You should see the prompt change to root@...:/workspace/U-MV#. You are now inside Linux; the commands of steps 9 to 13 are typed at this prompt and only here. ([+] Creating ... Network umv_default and Volume "umv_hf_cache" lines on the first start are normal.) Check the mounted folders:

```
ls /data
ls /weights
nvidia-smi
```

You should see ann_dir img_dir, then mambavision-b_generic-unet_acacia mambavision-s_generic-unet_acacia-88 mambavision-t_generic-unet_acacia, then the GPU table. *If /data or /weights prints nothing*: type exit and Enter (the prompt returns to PS D:\U-MV>), correct docker\.env (step 6), and repeat step 8.

Step 9 — Verify the software

inside the container (prompt ends with #) · once · needs Internet

```
python tools/verify_install.py --variant small
```

The first run downloads the MambaVision-S encoder (about 200 MB) from the Hugging Face Hub; progress bars and a warning that remote code was downloaded ("Make sure to double-check ...") are expected.

You should see every library with a version, a line CUDA available: True | <your GPU name> | torch cuda 11.8, mamba_ssm selective_scan_fn: ok, mmcv.ops (compiled extension): ok, U-MV-small: ... M parameters, forward (1, 3, 512, 512) -> (1, 2, 512, 512) in ... s, peak VRAM ... GiB, and the last line RESULT: OK. The last line RESULT: PROBLEMS FOUND (see above) means one of the lines above reports a problem: copy the whole output (right-click the title bar → Edit → Select All → Enter) into an

e-mail to the project lead.

Step 10 — Verify the dataset

inside the container · once

```
python tools/check_dataset.py /data --splits train val test2 Generalizability
```

You should see for each split a line such as [train] images=4893 masks=4893 pairs=4893 sidecars=0, then sampled ... sizes={(1024, 1024): ...} and mask values=[0, 1], and finally RESULT: OK. A note: ... side-car files ... are ignored line is harmless.

Step 11 — Identify the checkpoint

inside the container · once

```
python tools/inspect_checkpoint.py "/weights/mambavision-s_generic-unet_acacia-88"
```

You should see "path": ".../best_mIoU_iter_95000.pth", "variant": "small", "iteration": 95000 and => use configs/mambavision/U-MV-small.py with this checkpoint. *If red text says No best_mIoU_iter_*.pth ... found, /weights is wrong: exit, redo step 6, then step 8.*

Step 12 — Reproduce the published accuracy

inside the container · once · 20 to 30 minutes per run

```
CKPT="/weights/mambavision-s_generic-unet_acacia-88"
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split test2
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split Generalizability
```

The first line prints nothing (it stores the path in a name; it must be pasted again after every exit). Each run prints -> checkpoint loaded: ... no missing parameters, a progress line Iter(test) [100/3123] that advances, a small table with rows background and acacia, and one long last line starting with Iter(test) [3123/3123].

You should see in that last line, for test2, mIoU: 85.4 and mFscore: 91.6 (paper: 85.38 / 91.58), and for Generalizability, mIoU: 89.5 and mFscore: 94.2 (paper: 89.48 / 94.17), with the last digit possibly different. Send the two log files to the project lead; in Windows they are D:\U-MV\work_dirs\U-MV-small\test2\<date>\<date>.log and ... \Generalizability\<date>\<date>.log. The installation is now verified.

Optional, the other two models (paper: tiny 85.44 / 91.61, base 85.30 / 91.52 on test2):

```
CKPT_T="/weights/mambavision-t_generic-unet_acacia"
python tools/test.py configs/mambavision/U-MV-tiny.py "$CKPT_T" --test-split test2
CKPT_B="/weights/mambavision-b_generic-unet_acacia"
python tools/test.py configs/mambavision/U-MV-base.py "$CKPT_B" --test-split test2
```

Step 13 — Map an orthomosaic (optional)

inside the container · per orthomosaic

In Windows, open File Explorer → D: → A.tortilis_Data_Model → Data used to build the model, right-click an empty space → New → Folder, name it orthos, and copy the orthomosaic GeoTIFF into it. The file name must contain no spaces; site.tif is used below (replace it by the real name, twice). The container sees the file at once. Then, at the # prompt (the second command is written over four lines that end with a backslash; paste the four lines together and click **Paste anyway** if asked):

```
CKPT="/weights/mambavision-s_generic-unet_acacia-88"
python tools/geospatial_inference.py \
  --config configs/mambavision/U-MV-small.py --checkpoint "$CKPT" \
  --input /data/orthos/site.tif --output /data/predictions/site_crowns.gpkg \
  --scratch-dir /tmp/geospatial_work --min-area 1.0 --save-prob
```

You should see progress bars (tiles, writing prob, mean_prob) and a summary in curly braces with "polygons": <number>. In Windows the result is D:\A.tortilis_Data_Model\Data used to build the model\predictions\site_crowns.gpkg (and site_crowns_prob.tif); it opens in ArcGIS Pro or QGIS on top of the orthomosaic, with area (m²) and mean_prob attributes. For a folder of orthomosaics, use tools/batch_geospatial_inference.py with --input-dir /data/orthos --output-dir /data/predictions (see the inference chapter).

Step 14 — Leave and come back

both

At the # prompt, type exit and Enter: the container closes and the prompt returns to PS D:\U-MV>. Closing the PowerShell window does the same. Nothing is lost: the image stays built, the outputs are on D:.

Next time: start Docker Desktop and wait for the whale to be still; open Windows PowerShell (not as administrator); paste cd D:\U-MV and then docker\umv.cmd shell; you are back at the # prompt (step 8). To receive code updates first, paste at the PS prompt cd D:\U-MV, then \$env:GIT_LFS_SKIP_SMUDGE = "1", then git pull; the container sees the new files immediately.

Training on this workstation is possible for U-MV-tiny and U-MV-small with python tools/train.py configs/mambavision/U-MV-small.py --amp --cfg-options train_data_loader.batch_size=1 (12 GB GPU; the published runs used batch 2 on 24 GB); see the training chapter before starting a run.

If you see this message

First line of the message	Meaning	What to do
git : The term 'git' is not recognized or docker : The term 'docker' is not recognized	The program is not installed, or PowerShell was opened before it was installed	Install it (step 1); close PowerShell; open a new window; repeat the command
nvidia-smi : The term 'nvidia-smi' is not recognized	No NVIDIA driver, or no NVIDIA card in this computer	Check the first answer of 1a; if an NVIDIA card is listed, install its driver from nvidia.com and restart Windows; if none is listed, report it and stop
wsl : The term 'wsl' is not recognized	The Linux layer of Windows is not installed	Follow the administrator commands in step 1c; two restarts are needed
docker info shows a kernel ending in -linuxkit	Docker Desktop runs on Hyper-V, where no graphics card is visible	Docker Desktop → Settings → General → tick Use the WSL 2 based engine → Apply & restart
The term 'tools\windows\copy_archive.cmd' is not recognized or 'docker\umv.cmd' is not recognized	The prompt is not in the code folder	Paste cd D:\U-MV, then repeat the command
python : The term 'python' is not recognized	A container command was typed at the PS prompt	Do step 8 first, then repeat the command at the # prompt
Docker Desktop is not running or error during connect: ... dockerDesktopLinuxEngine	Docker Desktop is stopped	Start Docker Desktop, wait until the whale is still, repeat the command
no matching manifest for windows/amd64	Docker is in Windows-container mode	Right-click the whale icon → Switch to Linux containers , repeat the command

First line of the message	Meaning	What to do
could not select device driver "" with capabilities: [[gpu]] or Failed to initialize NVML	Docker cannot reach the GPU	Paste <code>wsl --update</code> , then <code>wsl --shutdown</code> ; start Docker Desktop again; repeat <code>docker\umv.cmd gpu</code> . Still failing: Docker Desktop → Settings → Resources → WSL integration → tick the default distribution → Apply & restart
nvidia-container-cli: initialization error: load library failed: libnvidia-ml.so.1 (usually with Auto-detected mode as 'legacy')	The Linux layer of Windows does not see the NVIDIA driver: its GPU support is outdated, or Docker Desktop is not using WSL 2. <code>docker\umv.cmd doctor</code> names the cause	1. <code>wsl --update</code> 2. quit Docker Desktop from the whale icon 3. <code>wsl --shutdown</code> 4. start Docker Desktop 5. repeat step 5. If it persists: Docker Desktop → Settings → General → tick Use the WSL 2 based engine → Apply & restart. If it still persists, install the current NVIDIA driver for the GPU from nvidia.com (choose the Studio or Game Ready driver, not a WSL driver, and install it on Windows, never inside WSL), restart Windows, repeat step 5
wsl --update: Invalid command line option or WSL not installed	The Windows version predates the WSL GPU support	The workstation needs Windows 11, or Windows 10 version 21H2 or newer (Windows key → type <code>winver</code>); report the version shown
Source not found: Z:\...	The share is not connected, or PowerShell runs as administrator	Open File Explorer → This PC and open Z:; open a normal PowerShell; repeat step 3
Clone succeeded, but checkout failed or This repository is over its data quota	Git downloaded the weight files	Paste <code>Remove-Item -Recurse -Force D:\U-MV</code> , then repeat step 2 from its first line
BUILD FAILED with cannot allocate memory above it	Docker has too little memory	Repeat step 4 with <code>memory=48GB</code> instead of <code>40GB</code> , then repeat step 7
BUILD FAILED (anything else)	A download was interrupted or a package failed	Repeat step 7 once; if it fails again, send the whole window content to the project lead
The argument 'D:\copy_archive.ps1' to the -File parameter does not exist	The copy helper of an older version had a fault	Paste <code>cd D:\U-MV</code> , then <code>git pull</code> , then repeat step 3
destination path 'U-MV' already exists and is not an empty directory	The old folder could not be deleted because a program still uses it	Close File Explorer, editors and any container window, then repeat step 2
docker\.env is missing	Step 3 did not write the file	Repeat step 3, or write the file by hand as shown at the end of step 6
Get-ChildItem : Access to the path ... is denied in step 3a	A protected folder was met while searching	Ignore it; if a folder line was printed, continue with 3b
ERROR: the checkpoint is U-MV-... but the config is U-MV-...	Config and checkpoint of different models	Use the config file named in the message
torch.OutOfMemoryError: CUDA out of memory or RuntimeError: CUDA out of memory	The GPU memory is used by something else	Close ArcGIS, QGIS and browsers; at the # prompt run <code>nvidia-smi</code> (Memory-Usage should be near 0 MiB); repeat the command. For step 13 add <code>--batch-size 2</code>

Reporting a problem

Give the step number, the exact command, and the whole window content (right-click the title bar → Edit → Select All → Enter, then paste into the e-mail). For steps 9 onward, also run at the # prompt

```
python tools/verify_install.py --variant small > work_dirs/verify.txt 2>&1
```

and attach the file D:\U-MV\work_dirs\verify.txt.

Contact: Mohamed Barakat A. Gibril (mbgibril@sharjah.ac.ae).

3 Framework overview

U-MV couples the hierarchical MambaVision encoder of Hatamizadeh and Kautz (2025), a hybrid of Mamba state-space mixers, self-attention and convolution, with a lightweight U-Net-style decoder. The encoder produces feature maps at strides 4, 8, 16 and 32; the decoder upsamples the deepest map stage by stage, concatenating skip features and applying two 3×3 convolution–BatchNorm–ReLU blocks per stage, and a 1×1 convolution yields two-class logits. Three encoder sizes are used: tiny (80/160/320/640 channels), small (96/192/384/768) and base (128/256/512/1024). Training minimises cross-entropy plus three times the Dice loss on 1024×1024 UAV tiles at 2.5 cm ground sampling distance.

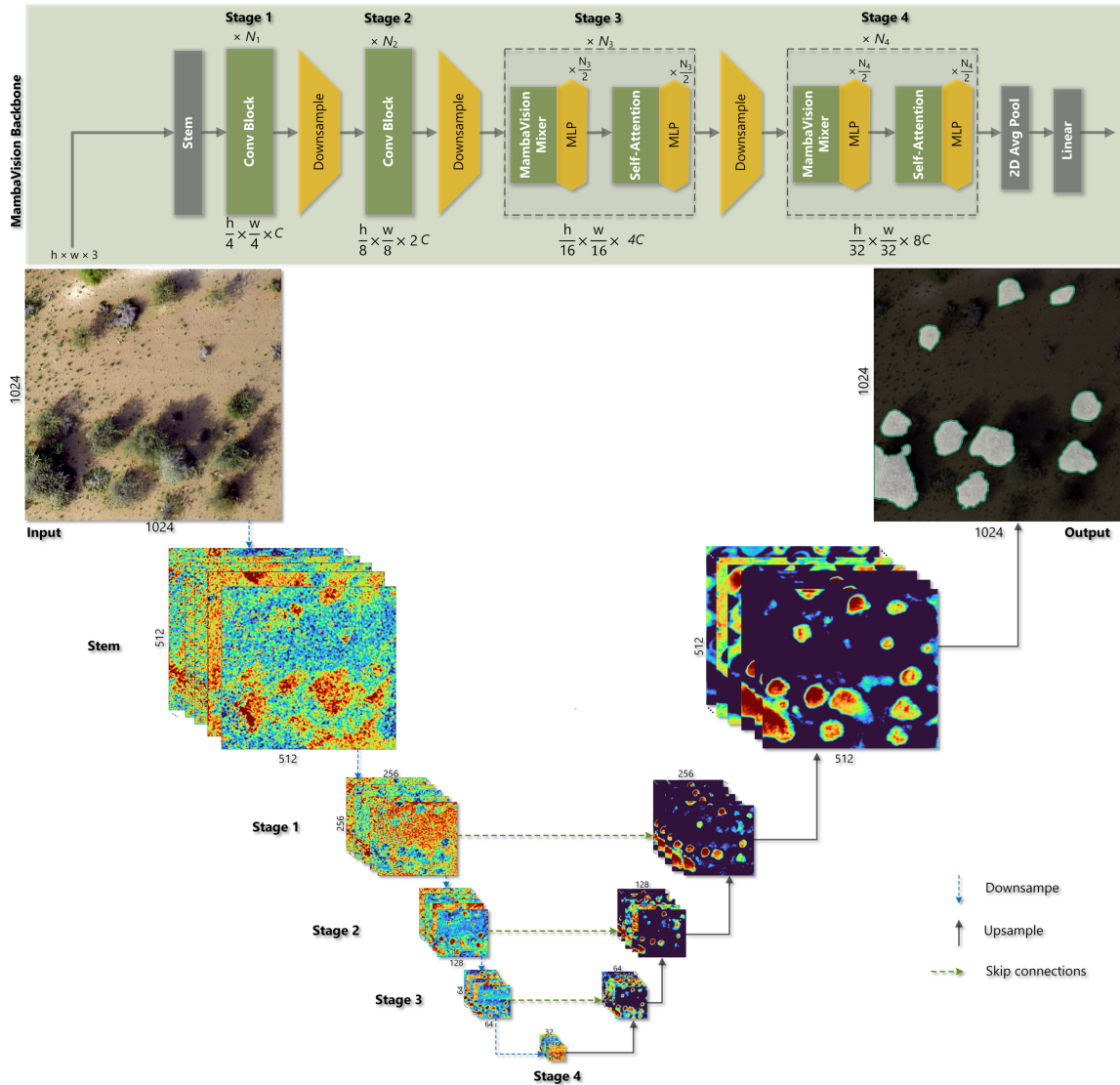


Figure 1. U-MV architecture: four-stage MambaVision encoder (top) and U-Net decoder with skip connections applied to a 1024×1024 UAV tile.

3.1 Highlights

- **Three variants** (tiny / small / base; 35.4 M and 54.2 M parameters for tiny and small) with released checkpoints; test-set mIoU 85.30–85.44 % and mF-score 91.52–91.61 %.
- **Self-contained MMSegmentation extension:** `pip install -e .` registers the backbone, decoder and dataset; no files are copied into MMSegmentation.

- **Reproducible environment:** Dockerfile pinned to the original stack (PyTorch 2.6.0 + CUDA 11.8, MMCV 2.1.0, MMSegmentation 1.2.2, mamba-ssm 2.2.4).
- **Regional inference:** seam-free sliding-window prediction over gigapixel orthomosaics, streaming vectorisation to GeoPackage/Shapefile with per-crown area and mean_prob attributes.

3.2 Repository layout

```

umv/
├── models/mamba_vision.py      installable package (registered with MMSegmentation)
├── models/unet_head.py        MambaVisionBackbone (Hugging Face nvidia/MambaVision-*-1K)
├── datasets/uav_acacia.py      GenericUNetHead (lightweight U-Net decoder)
├── inference/                  UAVAcaciaDataset (background / acacia)
├── checkpoint.py              tiling, blending, vectorisation, pipeline, CLI options
├──                             checkpoint inspection and variant detection
configs/
├── _base_/                     models/umv_unet.py · datasets/uav_acacia_dataset.py ·
├──                             schedules/schedule_100k_adamw.py · default_runtime.py
├── mambavision/                U-MV-tiny.py · U-MV-small.py · U-MV-base.py
tools/
├── train.py, test.py           MMSegmentation v1.2.2 entry points (+ --test-split)
├── geospatial_inference.py    one orthomosaic -> crown polygons
├── batch_geospatial_inference.py folder tree -> crown polygons
├── verify_install.py, inspect_checkpoint.py, download_backbones.py, check_dataset.py
docker/                          Dockerfile · docker-compose.yml · run.sh
requirements/                    core.txt · geo.txt · mamba.txt · dev.txt
Pretrained_Weights/              released checkpoints (Git LFS) + README
tests/                           CPU unit tests (tiling, vectorisation, pipeline)
docs/                            01 installation ... 08 hand-over checklist, REVISION_NOTES

```

3.3 Published accuracy

Model	Validation mIoU	Validation mF-score	Test mIoU	Test mF-score
U-MV-t	87.91	93.20	85.44	91.61
U-MV-s	88.02	93.27	85.38	91.58
U-MV-b	88.11	93.32	85.30	91.52

Values in percent (paper, Table 1). On the generalisability set (~11 km², 2 165 tiles) U-MV-s reached 89.48 % mIoU and 94.17 % mF-score. Training on a TITAN RTX (24 GB) with batch size 2 took 9.33 h (tiny), 11.8 h (small) and 18.12 h (base) for 100 000 iterations.

4 Software environment and installation

Two routes are supported. The Docker route reproduces the environment in which the published experiments were executed and is the recommended option for hand-over to new team members. The manual route documents every dependency for users who cannot run containers.

4.1 Software stack and version rationale

Component	Version	Reason
Python	3.11	Version of the original container (docs/reference/environment.frozen.yml).
PyTorch / torchvision	2.6.0 + cu118 / 0.21.0	Original experiments; CUDA 11.8 wheels run on driver ≥ 520 (TITAN RTX, RTX A5000).
MMEEngine	0.10.7	Version of the original container. Its checkpoint loader predates the <code>weights_only=True</code> default of PyTorch 2.6; umv registers a loader that reads the project checkpoints in full (and the image sets <code>TORCH_FORCE_NO_WEIGHTS_ONLY_LOAD=1</code>).
MMCV	2.1.0 (compiled)	Latest release accepted by MMSegmentation 1.2.2 (<code>mmcv>=2.0.0rc4,<2.2.0</code>). No prebuilt wheel exists for PyTorch 2.6, so it is compiled from source. The image compiles the operators for CPU by default (<code>MMCV_CUDA=0</code>): U-MV executes no MMCV CUDA operator, and the CPU build is GPU-independent and fast.
MMSegmentation	1.2.2	Last stable 1.x release; equivalent to the <code>main</code> branch used originally.
transformers / timm	4.50.0 / 1.0.15	Versions with which the MambaVision remote code (<code>trust_remote_code=True</code>) is known to import (<code>timm.models.registry</code> , <code>timm.models.layers.shims</code>).
mamba-ssm	2.2.4	Provides <code>selective_scan_fn</code> , the CUDA kernel used by the MambaVision mixer. Installed from the prebuilt wheel for torch 2.6 / CUDA 11 (kernels for compute capability 7.0 to 9.0); compiled from source only if the download fails.
GDAL / rasterio / geopandas	3.10 / 1.4 / 1.0 (conda-forge)	GDAL Polygonize is used for streaming vectorisation of gigapixel probability rasters.

MambaVision does **not** call `causal_conv1d`; that package is optional.

4.2 Docker route (recommended)

4.2.1 Prerequisites on Windows 11 / WSL2

- 1 Install an NVIDIA driver ≥ 520 on Windows (the driver is shared with WSL2; do not install a driver inside WSL2).
- 2 Install Docker Desktop with the WSL2 backend and enable the Ubuntu distribution under *Settings* → *Resources* → *WSL integration*.
- 3 Verify GPU visibility from WSL2:

```
docker run --rm --gpus all ubuntu:22.04 nvidia-smi --query-gpu=name,memory.total,compute_cap --format=csv
```

- 4 Grant WSL2 enough memory for the 1024×1024 training tiles (in `%USERPROFILE%\wslconfig`):

```
[wsl2]
memory=48GB
swap=16GB
```

then `wsl --shutdown` from PowerShell.

4.2.2 Build

```
git clone https://github.com/brakuta/U-MV-Acacia-tortilis-Crown-Mapping.git
cd U-MV-Acacia-tortilis-Crown-Mapping
git lfs install && git lfs pull          # released checkpoints; not needed with the project archive
cp docker/.env.example docker/.env && nano docker/.env # DATA_DIR, WEIGHTS_DIR
docker compose --env-file docker/.env -f docker/docker-compose.yml build
```

The build compiles MMCV (CPU operators, a few minutes) and installs the prebuilt mamba-ssm wheel; it typically takes 15–40 min and works unchanged on every supported NVIDIA GPU. Three build arguments exist: `MAX_JOBS` (parallel compiler jobs, default 4; each needs 2–4 GB of RAM, so use 2 where Docker/WSL2 has 16 GB or less), `MMCV_CUDA` (default 0; set 1 to compile MMCV's CUDA operators, which U-MV itself does not use) and, with `MMCV_CUDA=1`, `CUDA_ARCH` (default "7.0;8.0+PTX", which runs on every GPU generation from Volta to Ada; a single value such as 8.6 builds faster for that generation only):

```
docker compose --env-file docker/.env -f docker/docker-compose.yml build \
  --build-arg MAX_JOBS=2                # low-RAM host
docker compose --env-file docker/.env -f docker/docker-compose.yml build \
  --build-arg MMCV_CUDA=1 --build-arg CUDA_ARCH="8.6" # MMCV CUDA operators
```

`nvidia-smi --query-gpu=name,compute_cap --format=csv` prints the GPU name and its architecture number (7.5 = TITAN RTX / RTX 20xx, 8.6 = RTX A5000 / RTX 30xx, 8.0 = A100, 8.9 = RTX 40xx).

4.2.3 Windows (PowerShell + Docker Desktop)

`docker\umv.cmd` wraps the same compose commands for a PowerShell or cmd window, so that no long command has to be typed; `docker\.env.windows.example` is a ready-made `.env` with Windows paths (forward slashes, quoted):

```
copy docker\.env.windows.example docker\.env # then edit the two paths if needed
docker\umv.cmd gpu                          # Docker can see the GPU? (prints name, memory, arch)
docker\umv.cmd build                        # = compose build (MAX_JOBS=4)
docker\umv.cmd build 8.6 4                  # optional: MMCV CUDA operators for one architecture
docker\umv.cmd shell                        # = compose run --rm umv
```

The wrapper refuses to run when Docker Desktop is stopped or `docker\.env` is missing, and prints what to do.

The complete Windows procedure, including copying the archive from the share with `tools\windows\copy_archive.cmd`, is `docs\08_handover_checklist.md`.

4.2.4 Run

```
cp docker/.env.example docker/.env # set DATA_DIR and WEIGHTS_DIR (quoted if they contain spaces)
docker compose --env-file docker/.env -f docker/docker-compose.yml run --rm umv
# inside the container
python tools/verify_install.py --variant small
```

The compose file bind-mounts the repository at `/workspace/U-MV`, the dataset at `/data`, the checkpoint folder read-only at `/weights`, a named volume for the Hugging Face cache and `work_dirs/` for training outputs. `docker/run.sh` offers the same without compose and reads the same `docker/.env`.

4.2.5 Offline use

The MambaVision code and ImageNet weights are downloaded once from the Hugging Face Hub into the `hf_cache` volume:

```
python tools/download_backbones.py --variants tiny small base
export HF_HUB_OFFLINE=1 # subsequent runs need no network
```

4.3 Manual route (conda)

Install in this order; later steps compile against earlier ones.

```
conda create -n umv python=3.11 -y && conda activate umv
# 1. PyTorch (CUDA 11.8 wheels)
pip install torch==2.6.0 torchvision==0.21.0 --index-url https://download.pytorch.org/whl/cu118
# 2. MMEEngine and MMCV (compiled; requires nvcc from CUDA 11.8 toolkit on PATH)
pip install mmengine==0.10.7
MMCV_WITH_OPS=1 FORCE_CUDA=1 TORCH_CUDA_ARCH_LIST="7.5;8.6" pip install --no-build-isolation mmcv==2.1.0
# 3. Mamba kernels
pip install packaging ninja && pip install --no-build-isolation -r requirements/mamba.txt
# 4. Geospatial stack (GDAL bindings from conda-forge)
conda install -c conda-forge gdal=3.10 rasterio=1.4 geopandas=1.0 shapely=2.1 pyproj=3.7 pyogrio -y
# 5. Project
pip install -r requirements/core.txt
pip install --no-deps -e .
python tools/verify_install.py
```

`pip install -e .` makes `umv` importable from any working directory; the scripts in `tools/` also work without installation because they add the repository root to `sys.path`.

4.4 Verification

```
python tools/verify_install.py           # imports, CUDA, mamba_ssm, mmcv.ops
python tools/verify_install.py --variant small # builds U-MV-small, forward pass, peak VRAM
python -m pytest tests -q                # tiling / vectorisation unit tests (CPU)
```

Expected: RESULT: OK, and for the second command a forward pass of a 512×512 tensor producing logits of shape (1, 2, 512, 512).

5 Data

5.1 Source data

The framework was developed on RGB orthomosaics acquired with a senseFly eBee X (S.O.D.A. camera) at ~122 m AGL, processed in Pix4Dmapper to a ground sampling distance (GSD) of 2.5–3 cm. Annotations were produced with the semi-automated workflow described in Section 3.1 of the paper (SAM-LoRA crown proposals, Mask2Former–Swin candidate detection, visual verification).

5.2 Tiles and masks

Orthomosaics and rasterised crown polygons were partitioned into 1024×1024 pixel image–mask pairs. This tile size was chosen so that a mean crown (~17.5 m², ~5 m diameter, ~28 000 pixels at 2.5 cm) is embedded in sufficient context (shadows, terrain transitions, neighbouring land cover).

Requirements for new data:

Item	Specification
Images	3-band RGB, 8-bit, .tif (GeoTIFF or plain TIFF). Other suffixes: set <code>img_suffix</code> in the dataset config.
Masks	Single-band 8-bit index raster, 0 = background, 1 = acacia, .tif (or .png with <code>seg_map_suffix=' .png'</code>). Not RGB, not 0/255.
Names	Identical basenames for image and mask within a split.
Size	1024 × 1024 recommended; other sizes work (the pipeline resizes to 1024 on the long side and random-crops 1024 × 1024 during training).
Ignore value	255 in a mask is ignored by the loss (<code>seg_pad_val=255</code>).

Directory layout (mounted at /data in the container):

```
/data
├── img_dir/
│   ├── train/          4 893 pairs in the archive (26 615 used for the published models)
│   ├── val/            2 407
│   ├── test2/          3 123 (in-distribution test)
│   └── Generalizability/ 2 162 (out-of-distribution test, separate region)
└── ann_dir/
    ├── train/
    ├── val/
    ├── test2/
    └── Generalizability/
```

The archived training folder was pruned after the published runs (see the reproducibility chapter); the other three splits are complete.

A different root is selected without editing configs:

```
python tools/train.py configs/mambavision/U-MV-small.py \
  --cfg-options train_dataloader.dataset.data_root=/data/UAV_Acacia \
                 val_dataloader.dataset.data_root=/data/UAV_Acacia \
                 test_dataloader.dataset.data_root=/data/UAV_Acacia
```

5.3 Integrity check

```
python tools/check_dataset.py /data --splits train val test2 Generalizability
```

reports pair counts, tile sizes, dtypes, mask values and ArcGIS side-car files (*.tif.aux.xml, *.tif.ovr) per split. Side-cars are created when tiles are opened in ArcGIS Pro; the loader ignores them (only files ending in .tif are listed), but they can be deleted from working copies to save space.

A split whose folder is not named test (for example test2) is evaluated without renaming: `python tools/test.py <config> <ckpt> --test-split test2`.

5.3.1 Manual check

```
import numpy as np, rasterio, glob, os
root = '/data'
for split in ['train', 'val', 'test2', 'Generalizability']:
    imgs = sorted(glob.glob(f'{root}/img_dir/{split}/*.tif'))
    anns = sorted(glob.glob(f'{root}/ann_dir/{split}/*.tif'))
    assert [os.path.basename(p) for p in imgs] == [os.path.basename(p) for p in anns], split
    with rasterio.open(imgs[0]) as im, rasterio.open(anns[0]) as an:
        assert im.count == 3 and im.dtypes[0] == 'uint8', im.profile
        assert an.count == 1 and set(np.unique(an.read(1))) <= {0, 1, 255}, an.profile
        assert (im.width, im.height) == (an.width, an.height)
    print(split, len(imgs), 'pairs OK')
```

5.4 Export from ArcGIS Pro (summary)

- 1 Rasterise the verified crown polygons to the orthomosaic grid (Polygon to Raster, cell size = orthomosaic cell size, value field = 1, NoData → 0).
- 2 Export image and mask tiles with the same fishnet (Split Raster, tile size 1024×1024 , format TIFF, no overlap for training tiles).
- 3 Convert masks to 8-bit unsigned (Copy Raster, pixel type 8_BIT_UNSIGNED) and verify with the script above.

5.5 Location of the dataset

The dataset folder can reside anywhere; its path is supplied once in `docker/.env` (`DATA_DIR`) or on the command line (`--cfg-options train_data_loader.dataset.data_root=...`). Prefer a local SSD over a network share, because training performs ~30 000 random tile reads per epoch. When copying, ArcGIS side-cars can be excluded:

```
robocopy "<source>\Data used to build the model" "D:\UAV_Acacia_Data" /E /XF *.aux.xml *.ovr *.xml /MT:16
```

If a network drive must be read directly from WSL2, mount it first (`sudo mkdir -p /mnt/z && sudo mount -t drvfs Z: /mnt/z`) and quote the path.

5.6 Orthomosaics for regional inference

Inference operates on whole orthomosaics. Convert them to Cloud-Optimised GeoTIFF (COG) once; windowed reads are then efficient from any storage:

```
gdal_translate -of COG -co COMPRESS=DEFLATE -co BLOCKSIZE=512 -co NUM_THREADS=ALL_CPUS in.tif out_cog.tif
```

6 Training

6.1 Model and optimisation settings

All settings are declared in `configs/` and inherited by the three variant files; nothing is hard-coded in Python.

Setting	Value	Config location
Encoder	MambaVision-T/S/B, ImageNet-1K initialisation (Hugging Face Hub)	<code>_base_/models/umv_unet.py</code> → <code>model.backbone</code>
Encoder widths	T (80, 160, 320, 640) · S (96, 192, 384, 768) · B (128, 256, 512, 1024)	<code>mambavision/U-MV-*.py</code>
Decoder	U-Net style, widths (256, 128, 64, 32), BN + ReLU, bilinear ×2 upsampling	<code>model.decode_head</code>
Loss	CE + 3 × Dice (Eq. 1 of the paper)	<code>decode_head.loss_decode</code>
Optimiser	AdamW, lr 1e-4, $\beta = (0.9, 0.999)$, weight decay 0.05; backbone lr multiplier 0.1 (effective 1e-5); no decay on normalisation layers	<code>_base_/schedules/schedule_100k_adamw.py</code>
Schedule	100 000 iterations, polynomial decay (power 0.9), validation every 5 000 iterations	same
Gradient clipping	L2 norm 0.01	<code>optim_wrapper.clip_grad</code>
Batch	2 × 1024 × 1024 tiles	<code>_base_/datasets/uav_acacia_dataset.py</code>
Augmentation	RandomResize (0.5–2.0), RandomCrop 1024 (<code>cat_max_ratio=0.75</code>), horizontal flip, photometric distortion	same
Checkpointing	every 5 000 iterations, plus <code>best_mIoU_iter_*.pth</code> on validation mIoU	<code>default_hooks.checkpoint</code>
Test-time inference	sliding window 1024 × 1024, stride 512	<code>model.test_cfg</code>

All three variants share this schedule; the training logs of the released checkpoints confirm it (`docs/07_reproducibility.md`, §7.2). Note that the archived training split is a pruned subset (4 893 of the 26 615 tiles used for the published models), so retraining from the hand-over folder will not reach the published accuracy exactly.

6.2 Commands

```
# inside the container, dataset mounted at /data
python tools/train.py configs/mambavision/U-MV-tiny.py
python tools/train.py configs/mambavision/U-MV-small.py
python tools/train.py configs/mambavision/U-MV-base.py
```

Useful options (inherited from MMSegmentation):

Option	Effect
<code>--work-dir work_dirs/umv_small_run2</code>	Output directory (default <code>work_dirs/<config-name></code>).
<code>--resume</code>	Continue from the latest checkpoint in the work directory.
<code>--amp</code>	Automatic mixed precision (halves activation memory; not used for the published runs).
<code>--cfg-options train_data_loader.num_workers=16</code>	Override any config key.
<code>--cfg-options randomness.seed=0</code> <code>randomness.deterministic=True</code>	Fixed seed and deterministic cuDNN.

Outputs in the work directory: <timestamp>/<timestamp>.log, vis_data/scalars.json (loss and metric curves), iter_*.pth, best_mIoU_iter_*.pth, and a copy of the resolved configuration.

6.3 Expected resources (paper, TITAN RTX 24 GB, batch 2)

On a 12 GB GPU the published setting (batch 2, 1024×1024 crops, FP32) does not fit. Two configurations that keep the crop size and fit 12 GB are

```
python tools/train.py configs/mambavision/U-MV-small.py --amp \
    --cfg-options train_dataloader.batch_size=1 train_dataloader.num_workers=4
python tools/train.py configs/mambavision/U-MV-small.py --amp \
    --cfg-options train_dataloader.dataset.pipeline.3.crop_size="(512,512)" \
    model.data_preprocessor.size="(512,512)"
```

(batch 1 halves the samples seen per schedule; double max_iters to compensate. U-MV-base does not train reliably on 12 GB.) Evaluation and inference of all three variants fit 12 GB unchanged (test_dataloader.batch_size=1 is the default; the geospatial tool uses --batch-size 4 in FP16).

Variant	Parameters	FLOPs (1024 ²)	Training time (100k it.)	Validation mIoU
U-MV-t	35.41 M	0.144 T	9.33 h	87.91 %
U-MV-s	54.24 M	0.215 T	11.8 h	88.02 %
U-MV-b	not reported	0.396 T	18.12 h	88.11 %

6.4 Practical notes

- **First run downloads the backbone.** The MambaVision code and weights are fetched from the Hugging Face Hub (nvidia/MambaVision-*-1K) when the model is built. Pre-download with tools/download_backbones.py for offline hosts.
- **DataLoader workers.** The base config uses 8 workers. The original runs used 42 on a 64 GB workstation; increase train_dataloader.num_workers if the GPU is starved (check the data_time field in the log).
- **WSL2 memory.** Large Mamba backbones can trigger CUDA system-memory fallback and freeze WSL2 when host RAM is exhausted. Mitigations: disable the system fallback in the NVIDIA Control Panel (*CUDA – System Fallback Policy* → *Prefer No System Fallback*), keep .wslconfig memory below physical RAM, and drop the page cache between runs (sync; echo 3 > /proc/sys/vm/drop_caches from the WSL2 shell as root).
- **Monitoring.** tail -f work_dirs/<name>/<timestamp>/<timestamp>.log, or add vis_backends=[dict(type='LocalVisBackend'), dict(type='TensorboardVisBackend')] to configs/_base_/default_runtime.py (requires pip install tensorboard).

7 Evaluation

7.1 Metrics

IoUMetric with `iou_metrics=['mIoU', 'mFscore']` reports, per class and averaged over background and acacia, the intersection-over-union, F-score, precision and recall (Equations 2–7 of the paper). Averages are over the two classes (mIoU, mFscore); the acacia-class values (`IoU.acacia`, `Fscore.acacia`) are the quantities of ecological interest.

7.2 Commands

```
CKPT="/weights/mambavision-s_generic-unet_acacia-88" # folder -> its best_mIoU_iter_*.pth
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split test2 # test
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split Generalizability # 00D
python tools/test.py configs/mambavision/U-MV-small.py work_dirs/U-MV-small --test-split test2 # own run
# released checkpoints (identical to the best_mIoU files; after git lfs pull)
for v in tiny small base; do
  python tools/test.py configs/mambavision/U-MV-$v.py \
    Pretrained_Weights/U-MV-${v}_latest.pth --test-split test2
done
```

Without `--test-split` the config's default split `img_dir/test` is used (the archive names it `test2`).

Options:

Option	Effect
<code>--test-split NAME</code>	Evaluate on <code>img_dir/NAME</code> + <code>ann_dir/NAME</code> (added in this repository).
<code>--work-dir DIR</code>	Where the JSON metrics and log are written.
<code>--out DIR</code>	Save predicted index masks for offline analysis.
<code>--show-dir DIR</code>	Save colour overlays (input, ground truth, prediction).
<code>--tta</code>	Multi-scale (0.5–1.75) + horizontal-flip test-time augmentation (slower; not used in the paper).

Evaluation uses the config's sliding-window mode (1024×1024 window, stride 512).

7.3 Reference results (paper, Table 1)

Model	Validation mIoU	Validation mF-score	Test mIoU	Test mF-score
U-MV-t	87.91	93.20	85.44	91.61
U-MV-s	88.02	93.27	85.38	91.58
U-MV-b	88.11	93.32	85.30	91.52

Validation covers ~ 6.5 km² (2 407 archived tiles) and the independent test set ~ 6.7 km² (3 123 tiles, split `test2`). On the generalisability set (~ 11 km², 2 162 tiles) U-MV-s reached 89.48 % mIoU and 94.17 % mF-score (§4.4 of the paper). A freshly evaluated released checkpoint should reproduce the published values within about ± 0.1 percentage points; larger deviations indicate a data or preprocessing mismatch (band order, mask encoding, image suffix) rather than model drift.

To confirm that a checkpoint matches a config, `python tools/inspect_checkpoint.py <file or work dir>` prints the variant inferred from the decoder shapes together with the iteration and the MMEngine version stored in the checkpoint.

8 Regional inference on orthomosaics

`tools/geospatial_inference.py` (one image) and `tools/batch_geospatial_inference.py` (folder tree) share the implementation in `umv/inference/`. They turn a georeferenced orthomosaic into a GeoPackage (or Shapefile) of crown polygons with per-polygon attributes, without ever holding the full raster in memory.

8.1 Processing steps

- 1 **Staging (optional, `--scratch-dir`).** The input is copied to a fast local directory. This is recommended on WSL2/Docker when the data reside on a Windows drive (`/mnt/<drive>`), where random window reads are slow.
- 2 **Tiling.** An edge-aligned grid of `--tile-size` windows with `--overlap` pixels is generated; the last tile of each row/column ends exactly at the raster border, so every tile has full context.
- 3 **Prediction.** Tiles are normalised as in training (ImageNet mean/std, RGB) and passed through `EncoderDecoder.encode_decode`, yielding logits at tile resolution; the softmax probability of the acacia class is kept.
- 4 **Blending.** `--blend center` (default) writes only the region between the mid-points of neighbouring overlaps, so every pixel is predicted exactly once from the tile in which it lies farthest from the border (seam-free by construction). `--blend hann` averages all overlapping predictions with a 2-D Hann weight.
- 5 **Probability raster.** Accumulators are finalised into a tiled, compressed GeoTIFF in the CRS of the input (`--save-prob` keeps it; `--prob-dtype uint8` stores 0–255).
- 6 **Vectorisation.** The raster is thresholded at `--thresh`, polygonised with GDAL `Polygonize` (streaming, 4- or 8-connectivity), filtered by `--min-area` (CRS units, m² for projected CRS) and enriched with `area` and `mean_prob` (mean probability inside each polygon, for threshold tuning in GIS).

8.2 Commands

```
python tools/geospatial_inference.py \
  --config configs/mambavision/U-MV-small.py \
  --checkpoint Pretrained_Weights/U-MV-small_latest.pth \
  --input /data/orthos/Fujairah_block12_cog.tif \
  --output /data/predictions/Fujairah_block12_crowns.gpkg \
  --scratch-dir /tmp/geospatial_work --min-area 1.0 --save-prob
```

```
python tools/batch_geospatial_inference.py \
  --config configs/mambavision/U-MV-small.py \
  --checkpoint Pretrained_Weights/U-MV-small_latest.pth \
  --input-dir /data/orthos --output-dir /data/predictions \
  --scratch-dir /tmp/geospatial_work --skip-existing --continue-on-error
```

The batch script mirrors the input folder structure, skips finished outputs (`--skip-existing`, resumable) and writes `batch_summary.json`.

8.3 Parameters

Parameter	Default	Guidance
<code>--tile-size</code>	1024	Training tile size; keep.
<code>--overlap</code>	256	≥ 128 . Larger overlap = more context at the write boundary, more compute.
<code>--blend</code>	center	<code>hann</code> is marginally smoother on ambiguous crowns at a ~30 % higher cost.
<code>--batch-size</code>	8	16 fits on 24 GB GPUs in fp16 for U-MV-s; reduce on OOM.
<code>--precision</code>	fp16	CUDA autocast; fp32 for reference comparisons.

Parameter	Default	Guidance
<code>--thresh</code>	0.35	Probability threshold used for the regional maps. Tune per region with <code>mean_prob</code> ; 0.5 is the argmax-equivalent.
<code>--min-area</code>	0	e.g. <code>1 0</code> m ² removes spurious specks; the mean crown is ~17.5 m ² .
<code>--connectivity</code>	4	8 merges diagonally touching pixels.
<code>--bands</code>	1 2 3	Band indices fed as R, G, B (for 4-band RGBA orthomosaics <code>1 2 3</code> is still correct).
<code>--band-order</code>	rgb	Do not set <code>bgr</code> for standard orthomosaics; see below.
<code>--prob-dtype</code>	uint8	<code>float32</code> for full-precision probabilities (4× larger).
<code>--num-workers</code>	4	Raster reading processes.
<code>--mp-context</code>	(torch default)	<code>spawn</code> if forked workers misbehave with GDAL.

8.4 Channel order

During training, `LoadImageFromFile` reads TIFFs with OpenCV (BGR order) and `SegDataPreProcessor(bgr_to_rgb=True)` converts them to RGB before normalisation; the network therefore expects **RGB**. Rasterio returns bands in file order, which is RGB for standard orthomosaics, so the inference pipeline applies no channel swap. The previous versions of the scripts swapped the channels whenever `bgr_to_rgb=True` was set, which fed BGR tiles to the network; `--band-order bgr` reproduces that behaviour only for comparison.

8.5 Outputs

File	Content
<code><name>.gpkg</code> (layer <code><name></code>) or <code><name>.shp</code>	Crown polygons; attributes <code>area</code> (CRS units), <code>mean_prob</code> (0–1).
<code><name>_prob.tif</code> (with <code>--save-prob</code>)	Single-band probability raster; 0–255 for <code>uint8</code> .
<code>batch_summary.json</code> (batch)	Per-image tiles, polygons, timings, errors.

Post-processing in GIS: select polygons by `mean_prob`, dissolve touching crowns if instance separation is not required, and intersect with survey-zone boundaries.

8.6 Throughput

On a TITAN RTX with U-MV-s, fp16 and batch 8, throughput is roughly 4–6 tiles of 1024×1024 per second ($\approx 25\text{--}40 \text{ ha min}^{-1}$ at 2.5 cm GSD), dominated by raster I/O when reading from a Windows-mounted drive; staging to `/tmp` removes that bottleneck.

9 Reproducibility and paper-to-code mapping

9.1 Mapping of Section 3 of the paper to the configuration

Paper statement	Implementation
MambaVision T/S/B encoder, ImageNet-1K initialisation via timm/Hugging Face (§3.3, §3.5)	<code>MambaVisionBackbone(variant=..., pretrained=True) → AutoModel.from_pretrained('nvidia/MambaVision-{T,S,B}-1K', trust_remote_code=True)</code>
Encoder widths (80,160,320,640) / (96,192,384,768) / (128,256,512,1024) (§3.3)	<code>decode_head.encoder_channels</code> in <code>configs/mambavision/U-MV-*.py</code>
Lightweight CNN decoder with skip connections (§3.3)	<code>GenericUNetHead</code> , widths (256,128,64,32)
$L = L_{CE} + 3 L_{Dice}$ (Eq. 1)	<code>loss_decode=[CrossEntropyLoss(1.0), DiceLoss(3.0)]</code>
AdamW, $\beta=(0.9,0.999)$, weight decay 0.05 (§3.5)	<code>optimizer</code> in <code>_base_/schedules/schedule_100k_adamw.py</code>
Polynomial decay, power 0.9 (§3.5)	<code>PolyLR(power=0.9, eta_min=0)</code>
100 000 iterations, validation every 5 000 (§3.5, §4.2)	<code>train_cfg.val_interval=5000</code>
Batch size 2, tiles 1024×1024 (§3.2, §3.5)	<code>train_data_loader.batch_size=2, crop_size=(1024,1024)</code>
Gradient clipping (§3.5)	<code>clip_grad=dict(max_norm=0.01, norm_type=2)</code>
Random crop, horizontal flip, photometric perturbation (§3.5)	<code>train_pipeline</code>
Best validation checkpoint retained (§3.5)	<code>CheckpointHook(save_best='mIoU')</code>
Metrics: precision, recall, mIoU, mF-score (§3.6)	<code>IoUMetric(iou_metrics=['mIoU', 'mFscore'])</code>
Docker on a TITAN RTX workstation, 64 GB RAM (§3.5)	<code>docker/Dockerfile</code> (PyTorch 2.6.0 + CUDA 11.8 base); the original container definition is archived as <code>docs/reference/Dockerfile.original</code>

9.2 Reconciliation with the original training logs

The MMSegmentation work directories of the three published runs were recovered in September 2026 and their logs compared with the paper and the released configurations.

- Learning rate.** All three runs used a base learning rate of 1×10^{-4} with a backbone multiplier of 0.1; the logs list every encoder parameter at 1×10^{-5} . The paper's "initial learning rate of 1×10^{-5} " therefore refers to the encoder rate. The configs in this repository reproduce the logged setting.
- Schedule of the tiny variant.** The tiny log shows 100 000 iterations with validation every 5 000, identical to small and base, and reports the best validation mIoU (87.91 %, as in Table 1 of the paper) at iteration 100 000. The tiny config released earlier (lr 1×10^{-5} , 150 000 iterations) did not correspond to the run and has been aligned with the log.
- Checkpoints.** The released files are byte-identical (SHA-256) to `best_mIoU_iter_100000.pth` (tiny), `best_mIoU_iter_95000.pth` (small) and `best_mIoU_iter_60000.pth` (base) of the work directories, i.e. the best-validation checkpoints used for the paper's test results.
- Seeds.** No seed was fixed; runs are not bit-reproducible. For controlled comparisons add `--cfg-options randomness.seed=0 randomness.deterministic=True`.
- MMSegmentation revision.** The original container installed the main branch (`docs/reference/Dockerfile.original`); v1.2.2 is the last tagged release and is API-identical for the components used here.
- Channel order at inference.** The original inference scripts swapped channels to BGR before the network; the revised pipeline feeds RGB, which is what the training preprocessor produced

(docs/05_inference.md, §5.4). Regional maps produced with the old scripts may therefore differ slightly; tools/test.py is unaffected and provides the reference.

- 7 **Training split of the archived dataset.** The archived train folder holds 4 893 tiles (1024×1024 , identical in every copy found), whereas the paper reports 26 615. The tile indices run from 0 to 19 765 with gaps, and some files carry modification dates after the training runs (August 2025), which indicates that the training folder was pruned after the models were trained. Validation (2 407; the logs evaluate 1 204 batches of 2), test (3 123) and generalisability (2 162) tiles are complete. Consequently, evaluation and inference with the released checkpoints are fully reproducible from the archive, whereas retraining from the archive reproduces the recipe but not the published model.

9.3 What is and is not fixed by the pinned environment

Fixed: library versions, CUDA kernels, preprocessing, architecture, optimiser and schedule. Not fixed: cuDNN algorithm selection (cudnn_benchmark=True), data-loading order without a seed, and the exact Hugging Face revision of the MambaVision weights (pin a revision= in MambaVisionBackbone via model_name if the Hub repository is updated upstream).

9.4 Archival

The code is archived on Zenodo (DOI 10.5281/zenodo.18068379). When re-archiving this revision, include docs/reference/environment.frozen.yml, the Docker image digest (docker images --digests umv), and the SHA-256 of the checkpoints listed in Pretrained_Weights/README.md.

10 Troubleshooting

Symptom	Cause	Remedy
ModuleNotFoundError: No module named 'mmcv._ext'	mmcv-lite or a wheel built for another PyTorch/CUDA is installed.	Reinstall MMCV from source against the active PyTorch (docs/01_installation.md, step 2) or use the Docker image.
No module named 'mamba_ssm' / selective_scan_cuda import error	Kernels not built or built for a different PyTorch.	<code>pip install --no-build-isolation mamba-ssm==2.2.4</code> with nvcc available; confirm with <code>tools/verify_install.py</code> .
KeyError: 'MambaVisionBackbone is not in the mmseg::model registry'	umv not imported.	Configs must contain <code>custom_imports = dict(imports=['umv'])</code> (all shipped configs do); run scripts from the repository root or <code>pip install -e ..</code>
OSError: We couldn't connect to 'https://huggingface.co'	Backbone download blocked (proxy, offline).	Pre-download on a connected machine (tools/download_backbones.py), copy hf_cache/ and set HF_HUB_OFFLINE=1.
UnpicklingError: invalid load key, 'v' when loading a .pth	The file is a Git LFS pointer.	<code>git lfs install</code> & <code>git lfs pull</code> (see Pretrained_Weights/README.md).
_pickle.UnpicklingError: Weights only load failed	PyTorch ≥ 2.6 with an old MMEEngine.	Use MMEEngine 0.10.7 (pinned).
size mismatch for decode_head.decoder_stages.0.0.weight	Checkpoint and config variant differ.	<code>python tools/inspect_checkpoint.py <ckpt></code> prints the matching config.
Validation mIoU ≈ 50 % or all-background predictions	Masks are 0/255 or RGB, or wrong suffix.	Re-encode masks as 0/1 uint8 single band (docs/02_data_preparation.md, §2.3).
Test mIoU several points below the paper	Band order or resolution mismatch.	Ensure RGB orthomosaics, 2.5–3 cm GSD, no <code>--band-order bgr</code> .
CUDA out of memory during training	1024 ² tiles with batch 2 need ~20 GB on U-MV-b.	Use <code>--amp</code> , or <code>train_data_loader.batch_size=1</code> with doubled iterations.
WSL2 freezes; GPU memory shows 24 GB + system RAM use	CUDA system fallback exhausting host RAM.	Set <i>Prefer No System Fallback</i> in the NVIDIA Control Panel; reduce batch size; drop page cache between runs.
Docker build fails at the MMCV step with cannot allocate memory / Failed to build mmcv	too many parallel compiler jobs for the RAM given to Docker/WSL2	rebuild with <code>--build-arg MAX_JOBS=2</code> ; raise <code>memory=</code> in <code>.wslconfig</code> ; cached layers are reused
_pickle.UnpicklingError: Weights only load failed ... HistoryBuffer when a checkpoint is loaded	PyTorch ≥ 2.6 defaults <code>torch.load</code> to <code>weights_only=True</code> ; mmengine 0.10.x does not pass the argument	import <code>umv</code> before loading (all tools do; it registers a full loader) or set <code>TORCH_FORCE_NO_WEIGHTS_ONLY_LOAD=1</code> (set in the image)
ERROR: the checkpoint is U-MV-small but the config is U-MV-base	config and checkpoint belong to different variants	use the config named in the message; <code>tools/inspect_checkpoint.py</code> prints the variant
ERROR: N model parameters are not in the checkpoint	key layout mismatch (a foreign checkpoint, or a truncated download)	<code>tools/inspect_checkpoint.py <file> --keys</code> ; the released checkpoints have <code>backbone.backbone.model.* keys</code>
RuntimeError: CUDA error: no kernel image is available for execution on the device	MMCV CUDA operators (MMCV_CUDA=1) were compiled for another GPU generation	rebuild with the default <code>MMCV_CUDA=0</code> , or <code>CUDA_ARCH="7.0;8.0+PTX"</code>

Symptom	Cause	Remedy
AssertionError: The image size in a batch should be the same during tools/test.py	test tiles of different sizes with <code>batch_size > 1</code>	keep <code>test_dataloader.batch_size=1</code> (the default)
CUDA out of memory during tools/test.py or geospatial_inference.py on a 12 GB GPU	another process holds GPU memory, or a large batch	close GIS/browser applications; --batch-size 2 for the geospatial tool; --cfg-options test_cfg.fp16=True for test.py
docker\umv.cmd: Docker Desktop is not running	the Docker engine is stopped	start Docker Desktop and wait for the whale icon to be still
Clone succeeded, but checkout failed/over its data quota during git clone	Git LFS tried to download the released checkpoints	<code>\$env:GIT_LFS_SKIP_SMUDGE = "1"</code> before <code>git clone</code> ; use the archive's work directories
RuntimeError: DataLoader worker ... Bus error/ shared-memory errors in Docker	/dev/shm too small.	Run with --ipc=host (compose file does) or --shm-size=16g.
osgeo missing in inference	GDAL Python bindings absent.	Install via conda-forge; the rasterio fallback only handles rasters ≤ 1.5 gigapixels.
Straight cut lines in the crown map	--overlap below ~128 or --blend misconfigured.	Keep the defaults (overlap 256, centre-crop).
Very slow inference from /mnt/<drive>	Windows filesystem access from WSL2.	Use --scratch-dir /tmp/geospatial_work and COG inputs.
img_ratios NameError when loading a config	Legacy dataset config.	Fixed in this revision; <code>img_ratios</code> is defined in <code>configs/_base_/datasets/uav_acacia_dataset.py</code> .

A Revision notes (September 2026)

This revision prepares the repository for hand-over and independent reproduction. The network architecture, released configurations and checkpoints are unchanged; the changes concern packaging, correctness of the auxiliary code, portability and documentation.

A.1 Defects found in the previous revision

#	Defect	Effect	Resolution
1	<code>mmseg/dataset/ade.py</code> (singular <code>dataset</code>) was meant to overwrite <code>MMSegmentation</code> 's <code>mmseg/datasets/ade.py</code> ; the copy-in instructions never placed it there.	Configs referenced <code>ADE20KDataset</code> , which expects <code>.jpg</code> images and 150 classes; a fresh installation could not load <code>.tif</code> tiles or report two-class metrics.	New registered dataset <code>UAVAcaciaDataset</code> in <code>umv/datasets/</code> ; no upstream file is modified.
2	<code>img_ratios</code> undefined in the dataset config.	<code>Config.fromfile</code> raised <code>NameError</code> on every config.	Defined.
3	<code>configs/_base_/default_runtime.py</code> absent from the repository.	Configs unloadable outside a full <code>MMSegmentation</code> checkout.	Added (verbatim from v1.2.2).
4	Inference scripts swapped <code>RGB</code> → <code>BGR</code> whenever <code>bgr_to_rgb=True</code> .	Network received <code>BGR</code> tiles although it was trained on <code>RGB</code> .	Removed; <code>--band-order bgr</code> retained for comparison.
5	Batch script used <code>mode='whole'</code> with <code>ori_shape=(h, w)</code> for border tiles.	Padded border tiles were resized to the valid size, misaligning predictions along raster edges.	Both scripts now use <code>encode_decode</code> at tile resolution and crop.
6	Centre-crop writer double-wrote border regions when a raster edge fell inside the first tile of a row.	Silent averaging at borders.	Edge-aligned grid with mid-point write boundaries; property-tested (<code>tests/test_tiling.py</code>).
7	Minimum-area filtering documented in the README but not implemented.	—	<code>--min-area</code> implemented; <code>area</code> attribute added.
8	Hard-coded paths, thresholds and device settings inside the scripts.	Edits required for every run.	<code>argparse</code> CLI with documented defaults.
9	<code>num_workers=42</code> in the dataset config.	Failure or thrashing on smaller hosts.	8 (documented).
10	Backbone always downloaded ImageNet weights, even when loading a fine-tuned checkpoint; no offline path.	Network access mandatory at inference time.	<code>pretrained=False</code> when a checkpoint is given; <code>HF_HUB_OFFLINE</code> / <code>local_files_only</code> supported; <code>tools/download_backbones.py</code> .
11	<code>environment.yml</code> was a full conda dump of the container's base environment (prefix: <code>/opt/conda</code> , 300 packages).	Not recreatable; misleading as a specification.	Moved to <code>docs/reference/environment.frozen.yml</code> ; replaced by <code>requirements/*.txt</code> , <code>pyproject.toml</code> and <code>docker/Dockerfile</code> .
12	<code>Assets/mamba_vision_architecture.png</code> was 92 MB ($14\,639 \times 14\,198$ px).	Slow clones, README rendering issues.	Downscaled to 2 400 px (2.5 MB); the original remains in Git history.
13	<code>.idea/workspace.xml</code> committed.	IDE noise.	Removed and ignored.
14	<code>torch==4.65.2</code> , <code>torch==2.6.0+cu118</code> pins in <code>requirements.txt</code> not installable from PyPI without extra index.	<code>pip install -r requirements.txt</code> failed.	Split requirements; <code>PyTorch/MMCV/mamba-ssm</code> installed explicitly.

A.2 Structural changes

- `umv/` installable package (`pip install -e .`) with `models/`, `datasets/`, `inference/`, `checkpoint.py`. Registered names `GenericUNetHead`, `mamba_{tiny,small,base}_vision_timm` are kept, and `MambaVisionBackbone` is added, so configs embedded in old checkpoints still resolve.
- Module attribute names (`backbone.backbone.*`, `decode_head.decoder_stages.*`, `decode_head.segmentation_head.*`, `decode_head.conv_seg.*`) are preserved; released checkpoints load without key remapping.
- Configs split into `_base_/models`, `_base_/datasets`, `_base_/schedules`, `_base_/default_runtime.py`; variant files contain only variant-specific fields. Numerical hyper-parameters are unchanged.
- `tools/train.py` and `tools/test.py` vendored from `MMSegmentation v1.2.2` (plus `--test-split`), so the repository is self-contained.
- New tools: `verify_install.py`, `inspect_checkpoint.py`, `download_backbones.py`, `batch_geospatial_inference.py` (renamed from `Batch_processing_geospatial_inference.py`).
- `tests/` (14 tests) covering tiling, blending, vectorisation, checkpoint variant detection and the end-to-end inference pipeline on synthetic GeoTIFFs (CPU).

A.3 Compatibility with the archived work directories

`umv/compat.py::load_config` detects configs that import `mmseg.custom_models.*` (the archived training configs) and maps them to the new package, including `ADE20KDataset` → `UAVAcaciaDataset`. All tools use it, so `tools/test.py` `<archived config>` `<archived checkpoint>` works without edits. `docker/.env` (`DATA_DIR`, `WEIGHTS_DIR`) decouples the container from the location of the hand-over folder.

A.4 Validation performed in this revision

- All three configs load; models build with a stub encoder and run a training step (CE + Dice losses), `encode_decode`, and sliding-window prediction on CPU (MMEEngine 0.10.7, MMSegmentation 1.2.2).
- The inference pipeline recovers synthetic crowns with correct CRS, areas ($\pm 5\%$), attributes and seam-free tiling for both blending modes.
- The Git LFS objects of the three released checkpoints are present on GitHub and their SHA-256 values equal those of `best_mIoU_iter_{100000,95000,60000}.pth` in the `tiny/small/base` work directories.
- The original container definition was recovered and archived as `docs/reference/Dockerfile.original`; it uses the same base image and MMCV build route as `docker/Dockerfile`.

A.5 Fixes after the first installation attempt (September 2026)

A review of the hand-over material against the receiving workstation (Windows 11, Docker Desktop, 64 GB RAM, 12 GB GPU) led to the following corrections:

- `umv` registers an `mmengine` checkpoint loader that passes `weights_only=False`; under PyTorch 2.6 the stock loader rejects the `message_hub` objects in the checkpoints (`tools/test.py`, `init_model`).
- `tools/test.py` builds the backbone with `pretrained=False`, refuses a config/checkpoint variant mismatch and fails on missing parameters instead of evaluating a partly random network.
- `test_dataloader/val_dataloader` batch size 1 (fits 12 GB, avoids the same-size-in-batch assertion); `geospatial --batch-size` default 4.
- Inference pipeline: the scratch directory is never deleted itself (only a private sub-folder), the probability raster is always GeoTIFF, `.vrt` inputs are rejected when staging; vectorisation handles inputs without CRS and computes areas in metres for geographic CRSs.

- Docker: mmcv operators compiled for CPU by default (GPU-independent image, minutes instead of tens of minutes), mamba-ssm from its prebuilt wheel, setuptools<82 pin, correct compose fallbacks, TORCH_FORCE_NO_WEIGHTS_ONLY_LOAD.
- Windows helpers: docker\umv.cmd checks that Docker Desktop is running, reports the GPU architecture, and builds without arguments; tools\windows\copy_archive.cmd copies the archive and checks it; the procedure uses GIT_LFS_SKIP_SMUDGE=1 for git clone, writes .wslconfig from PowerShell (40 GB) and is written for a first-time user.

A.6 Not validated here (requires a GPU host)

- The Docker image build (MMCV and mamba-ssm compilation) and the Hugging Face backbone instantiation; the Hub was unreachable from the sandbox in which this revision was prepared. Run docs/08_handover_checklist.md steps 7–12 on the workstation and report any deviation.
- Numerical equivalence of the released checkpoints with the paper's metrics.

A.7 Resolved from the recovered work directories (September 2026)

- The tiny configuration has been aligned with its training log (lr 1e-4, 100 000 iterations); the previously released tiny config did not match the run.
- The released checkpoints are the best-validation checkpoints (SHA-256 verified against best_mIoU_iter_{100000,95000,60000}.pth).
- The archived training split (4 893 tiles) is a pruned subset of the 26 615 tiles used for the published models; no fuller copy exists on the project disks. Evaluation and inference are exact; retraining reproduces the recipe.

A.8 Decisions left to the authors

- 1 Whether to publish the checkpoints on Zenodo/Hugging Face in addition to Git LFS (bandwidth quota of GitHub LFS is 1 GB month⁻¹ on free plans; three full downloads exhaust it).
- 2 Whether regional maps produced with the earlier BGR-swapped scripts should be regenerated.

B Pretrained weights

The three released checkpoints are stored with **Git LFS**. A plain `git clone` without LFS yields 134-byte pointer files, not weights. Fetch the binaries with:

```
git lfs install
git lfs pull # all three (~815 MB)
git lfs pull --include="Pretrained_Weights/U-MV-small_latest.pth" # one variant
```

Variant	File	Size	SHA-256 (LFS oid)
U-MV-tiny	U-MV-tiny_latest.pth	159 MB	85796b37582647a6d83973e240ebf30e0a562676bd44ae84eb81c953dc8900ac
U-MV-small	U-MV-small_latest.pth	234 MB	0c8d2b3dcd6f6272ea12d9fcdac2cba79ca185277cb2f5d1eff55e5d065ed9d8
U-MV-base	U-MV-base_latest.pth	422 MB	fcd86b17fb1c49e4f9555538526cc0edff805d0303463bbcc9336b7e537b66d3

Verify a download with `sha256sum <file>` and identify the variant of any local checkpoint with:

```
python tools/inspect_checkpoint.py /path/to/checkpoint.pth
```

A checkpoint obtained elsewhere (e.g. a `best_mIoU_iter_*.pth` from a `work_dirs/` folder) can be used directly by passing its path to `tools/test.py` or the inference scripts; no renaming is required.

B.1 Provenance (verified September 2026)

The released files are byte-identical to the best-validation checkpoints of the original MMSegmentation work directories (SHA-256 verified):

Released file	Work directory	Checkpoint	Iteration
U-MV-tiny_latest.pth	mambavision-t_generic-unet_acacia	best_mIoU_iter_100000.pth	100 000
U-MV-small_latest.pth	mambavision-s_generic-unet_acacia-88	best_mIoU_iter_95000.pth	95 000
U-MV-base_latest.pth	mambavision-b_generic-unet_acacia	best_mIoU_iter_60000.pth	60 000

The `_latest` suffix therefore denotes the latest release, not the last training iteration. `iter_100000.pth` files in the work directories are the final-iteration checkpoints and were not released.

B.2 Original work directories

Passing a work-directory folder to any tool selects its `best_mIoU_iter_*.pth`:

```
python tools/inspect_checkpoint.py "/weights/mambavision-s_generic-unet_acacia-88"
python tools/test.py configs/mambavision/U-MV-small.py \
    "/weights/mambavision-s_generic-unet_acacia-88" --test-split test2
```

C Command reference

C.1 Windows (PowerShell, in D:\U-MV)

```
$env:GIT_LFS_SKIP_SMUDGE = "1" # before git clone / git pull (no LFS download)
tools\windows\copy_archive.cmd D:\A.tortilis_Data_Model # copy dataset + weights from Z:, check counts
copy docker\.env.windows.example docker\.env # then edit the two paths if needed
docker\umv.cmd gpu # GPU visible to Docker? (name, memory, arch)
docker\umv.cmd build # image build (all GPUs); "build 8.6 4" = MMCV CUDA ops
docker\umv.cmd shell # interactive container; exit to leave
```

C.2 Environment (Linux / WSL2)

```
cp docker/.env.example docker/.env # set DATA_DIR and WEIGHTS_DIR
docker compose --env-file docker/.env -f docker/docker-compose.yml build
docker compose --env-file docker/.env -f docker/docker-compose.yml run --rm umv
python tools/verify_install.py --variant small # stack check + forward pass
python tools/download_backbones.py # cache backbones; then export HF_HUB_OFFLINE=1
python -m pytest tests -q # unit tests (CPU)
```

C.3 Data and checkpoints

```
python tools/check_dataset.py /data --splits train val test2 Generalizability
python tools/inspect_checkpoint.py "/weights/mambavision-s_generic-unet_acacia-88"
```

C.4 Training

```
python tools/train.py configs/mambavision/U-MV-small.py
python tools/train.py configs/mambavision/U-MV-small.py --resume --work-dir work_dirs/U-MV-small
python tools/train.py configs/mambavision/U-MV-small.py \
  --cfg-options randomness.seed=0 randomness.deterministic=True
```

C.5 Evaluation

```
CKPT="/weights/mambavision-s_generic-unet_acacia-88" # folder resolves to best_mIoU_iter_*.pth
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split test2
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split Generalizability
python tools/test.py configs/mambavision/U-MV-small.py "$CKPT" --test-split test2 \
  --out preds/ --show-dir vis/
```

C.6 Regional inference

```
CKPT="/weights/mambavision-s_generic-unet_acacia-88"
python tools/geospatial_inference.py \
  --config configs/mambavision/U-MV-small.py --checkpoint "$CKPT" \
  --input /data/orthos/block12_cog.tif \
  --output /data/predictions/block12_crowns.gpkg \
  --scratch-dir /tmp/geospatial_work --min-area 1.0 --save-prob

python tools/batch_geospatial_inference.py \
  --config configs/mambavision/U-MV-small.py --checkpoint "$CKPT" \
  --input-dir /data/orthos --output-dir /data/predictions \
  --scratch-dir /tmp/geospatial_work --skip-existing --continue-on-error
```

C.7 Orthomosaic preparation

```
gdal_translate -of COG -co COMPRESS=DEFLATE -co BLOCKSIZE=512 \
  -co NUM_THREADS=ALL_CPUS in.tif out_cog.tif
```